

Telescoping Solutions of Differential Equations: A High-Order Exponential Approximation Framework

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Abstract

We construct high-order approximations to solutions of linear ordinary differential equations (ODEs) by replacing the exponentials in their closed-form solutions with *telescoping exponential approximants* $E_n^{(s)}(c) = (1 + c/n + s\frac{\sqrt{3}}{6}c^2n^{-2})^{n+c(1/2-s\sqrt{3}/6)}$, $s = \pm 1$, whose successive differences decay as $O(n^{-4})$. We give explicit order-3 constructions for first-order linear ODEs $y' = ay$, the harmonic oscillator $y'' + \omega^2 y = 0$, and matrix systems $x' = Ax$; a general theorem showing that any constant-coefficient linear ODE whose solution is a finite combination of $t^j e^{\lambda t}$ inherits telescoping sequences of arbitrary order; and error analysis giving uniform bounds $O(n^{-(k+1)})$ on differences and $O(n^{-k})$ on errors over compact time intervals. The exact algebraic identities the framework rests on — the vanishing of the n^{-1} and n^{-2} log-series coefficients, the leading coefficient g_3 , the finite-difference factor -3 , and the Pythagorean base identity — are kernel-checked in Lean 4 (Appendix 8); the asymptotic statements themselves are confirmed by high-precision computation and are *not* claimed as formalised. Every claim is labelled by evidence tier.

1 Introduction

The approximation of solutions to differential equations is a central problem in computational mathematics. Classical numerical schemes (Runge–Kutta, linear multistep) deliver discrete approximations without an explicit analytical error structure [8]. Exponential integrators [16, 15] exploit the exponential structure of linear problems but require machinery for matrix functions [1, 17].

A convergent sequence $(x_n) \rightarrow L$ satisfies the telescoping identity $L = x_1 + \sum_{n \geq 1} (x_{n+1} - x_n)$, so the convergence rate is the decay rate of the differences $D_n = x_{n+1} - x_n$. Accelerating such convergence by transforming a given sequence is a classical subject with a large literature (surveys: [20, 7]; standard reference [12, §3.9]). Rather than transform a given sequence, this note *constructs* sequences whose differences decay quickly and lifts them from constants to functions and to ODE solutions.

Distinction from symbolic “creative telescoping.” Creative telescoping [10, 6] finds differential or difference equations satisfied by special functions — a conceptually different problem. Our use of “telescoping” refers to the classical analytic telescoping of series, and to the *numerical* acceleration context of [20, 7].

Prior work. The order-3 approximant used here as a building block was introduced and analysed in the author's earlier work [4] (which proves the $O(n^{-4})$ telescoping decay of this family through parameter optimisation); the function-lifting methodology (logarithms, trigonometric values) is developed in [5]. The present paper applies these to differential equations.

2 Preliminaries

Definition 1 (Telescoping order). A sequence (x_n) has *telescoping order* k if $D_n := x_{n+1} - x_n = O(n^{-(k+1)})$. Then the tail $R_N = \sum_{n>N} D_n = O(N^{-k})$, and in particular $|L - x_N| = O(N^{-k})$.

Definition 2 (Uniform telescoping). A sequence of functions (f_n) on an interval I has *uniform telescoping order* k on I if $f_n \rightarrow f$ uniformly and $\sup_{t \in I} |D_n(t)| = O(n^{-(k+1)})$.

3 The exponential engine

We use the *c-in-base* parametrisation (the base carries a factor of c ; this differs from the parametrisation of [5] and the choice is what makes the cancellations below work):

Definition 3 (Order-3 exponential approximant). For $s \in \{+1, -1\}$ and $n \geq 1$, set $\alpha_s(c) = s \frac{\sqrt{3}}{6} c^2$, $\beta_s(c) = c(\frac{1}{2} - s \frac{\sqrt{3}}{6})$, and

$$E_n^{(s)}(c) = \left(1 + \frac{c}{n} + \frac{\alpha_s(c)}{n^2}\right)^{n+\beta_s(c)}.$$

The base carries no factor of c in the exponent; c enters the base through c and α_s , and the exponent through β_s . For $|c|$ bounded and n large, the base lies in a neighbourhood of 1 avoiding $(-\infty, 0]$, so the power is the principal branch and is analytic in c ; the expansion below is uniform on compact c -sets for $n \geq n_0$. (Throughout, “ $\log E_n$ ” denotes the analytic exponent $M(u) \log B(u)$ below, i.e. $\log E_n^{(s)}(c) = (u^{-1} + \beta_s(c)) \log(1 + cu + \alpha_s(c)u^2)$ with $u = 1/n$; this is the analytic logarithm of E_n near $n = \infty$, not the principal logarithm of the complex number $E_n^{(s)}(c)$.)

Theorem 4 (Exponential telescoping engine). For each $c \in \mathbb{C}$, with $E_n^{(s)}$ as in Definition 3:

1. $E_n^{(s)}(c) \rightarrow e^c$ as $n \rightarrow \infty$; *[num, classical]*
2. $E_{n+1}^{(s)}(c) - E_n^{(s)}(c) = O(n^{-4})$; *[num]*
3. convergence is uniform on compact subsets of $\mathbb{C}$; *[classical]*
4. writing $u = 1/n$, the analytic exponent has the expansion

$$\log E_n^{(s)}(c) - c = g_3^{(s)}(c) n^{-3} + O(n^{-4}), \quad g_3^{(s)}(c) = \frac{c^4(-3 + 2s\sqrt{3})}{72},$$

with $g_4^{(s)}(c) = c^5(39 - 25s\sqrt{3})/720$, $g_5^{(s)}(c) = c^6(-18 + 11s\sqrt{3})/270$. *[Lean] (the g_3 identities: u^1, u^2 vanish, $u^3 = g_3$), [num] (g_4, g_5 and the asymptotics)*

Derivation of the cancellation. Write $u = 1/n$, $B(u) = 1 + cu + \alpha_s u^2$, $M(u) = u^{-1} + \beta_s$, so $\log E_n^{(s)}(c) = M(u) \log B(u)$. Since $B(0) = 1$ and $\log(1 + z) = z - z^2/2 + z^3/3 - \dots$,

$$\log B(u) = cu + \left(\alpha_s - \frac{c^2}{2}\right)u^2 + \left(\frac{c^3}{3} - c\alpha_s\right)u^3 + O(u^4).$$

Multiplying by $M(u) = u^{-1} + \beta_s$ and subtracting c gives coefficient expressions in u ; the u^1 coefficient is

$$A_1 = \left(\alpha_s - \frac{c^2}{2}\right) + \beta_s c,$$

and the u^2 coefficient is

$$A_2 = \left(\frac{c^3}{3} - c\alpha_s\right) + \beta_s \left(\alpha_s - \frac{c^2}{2}\right) = -\frac{c^3}{6} + \beta_s c^2 - \beta_s^2 c.$$

One checks directly that $A_1 = 0$ identically and $A_2 = -c^3((s\sqrt{3})^2 - 3)/36 = 0$ when $(s\sqrt{3})^2 = 3$; the u^3 coefficient is then $A_3 = \left(-\frac{\alpha_s^2}{2} + c^2\alpha_s - \frac{c^4}{4}\right) + \beta_s \left(\frac{c^3}{3} - c\alpha_s\right) = g_3^{(s)}(c)$. These three polynomial identities are kernel-checked (Appendix 8, `u1_coeff_zero`, `u2_coeff_zero`, `u3_coeff_eq_g3`); the identification of A_1, A_2, A_3 as the u^1, u^2, u^3 coefficients uses the classical logarithm series. Uniformity on compact c -sets is standard; convergence to e^c and the $O(n^{-4})$ rate then follow from $E_n^{(s)}(c) = e^c(1 + g_3^{(s)}n^{-3} + O(n^{-4}))$ and the finite-difference factor below. $\square$

Lemma 5 (Finite-difference factor). *For a term behaving as $u^3 = n^{-3}$, the successive difference carries the factor -3 : exactly, $\left(\frac{u}{1+u}\right)^3 - u^3 = (-3u^4 - 3u^5 - u^6)/(1+u)^3$, whose leading coefficient is $-3u^4$. Hence*

$$E_{n+1}^{(s)}(c) - E_n^{(s)}(c) = -3g_3^{(s)}(c)e^c n^{-4} + O(n^{-5}) = \frac{c^4(3 - 2s\sqrt{3})}{24} e^c n^{-4} + O(n^{-5}).$$

Proof. The rational identity and the coefficient -3 are kernel-checked (Appendix 8, `diff_cube_factor`); the leading-constant identification $-3g_3^{(s)} = c^4(3 - 2s\sqrt{3})/24$ is `lead.eq`. The $O(n^{-5})$ remainder uses that $\log E_n^{(s)} - c$ is real-analytic in u at 0 (the base is smooth and positive near $u = 0$), so the expansion is a convergent power series whose $O(u^5)$ truncation error has an $O(u^5)$ successive difference. $\square$

Remark 6 ($O(\cdot)$ is a bound, not a sharp law). The estimates $N = O(\varepsilon^{-1/3})$ follow from $R_N = O(N^{-3})$ as an upper bound and become a sharp power law only when the leading constant is nonzero. At degenerate parameters (e.g. $c = 0$) the effective order is higher.

4 First-order linear ODEs

Consider $y' = ay$, $y(0) = y_0$, with exact solution $y(t) = y_0 e^{at}$. Define $y_n(t) := y_0 E_n^{(s)}(at)$.

Theorem 7 (First-order convergence). *For any compact interval $I \subset \mathbb{R}$: $\sup_{t \in I} |y_n(t) - y(t)| = O(n^{-3})$ and $\sup_{t \in I} |y_{n+1}(t) - y_n(t)| = O(n^{-4})$.* **[num, classical]**

Corollary 8 (Asymptotic constants). *Write $\tilde{A}_3(c) := g_3^{(s)}(c) = c^4(-3 + 2s\sqrt{3})/72$, so that $E_n^{(s)}(c) - e^c = e^c(\tilde{A}_3(c)n^{-3} + O(n^{-4}))$: the coefficient of n^{-3} in the normalised error $e^{-c}(E_n^{(s)}(c) - e^c)$ is $\tilde{A}_3$, and in the actual error it is $e^c \tilde{A}_3(c)$. Then for each compact $I = [-T, T]$ there is $n_0 = n_0(a, T)$ and a constant C_1 such that, for all $n \geq n_0$,*

$$\sup_{t \in I} |y(t) - y_n(t)| \leq C_1 n^{-3}, \quad \sup_{t \in I} |y_{n+1}(t) - y_n(t)| \leq 3C_1 n^{-4},$$

where one may take $C_1 = |y_0| e^{|a|T} (\sup_{c \in [-|a|T, |a|T]} |\tilde{A}_3(c)| + 1)$.

Proof. The factor 3 in the second bound is Lemma 5; the $e^{|a|T}$ factor absorbs the e^c normalisation. These are *asymptotic* bounds: they hold for $n \geq n_0$, and the “+1” absorbs the $O(n^{-4})$ tail so that no unproven finite- n inequality is asserted. (Leading coefficients alone do not give an exact global constant; the statement is deliberately an inequality valid for large n .) $\square$

Remark 9 (Complexity, stated honestly). To reach accuracy ε one needs $n = O(\varepsilon^{-1/3})$ terms. This exponent is *smaller* than the $\varepsilon^{-1/2}$ of a method whose error decays as $O(N^{-2})$ [20, 7], but term count is not a work model: evaluating $E_n^{(s)}(at)$ is a single non-integer power, and against analytic targets on bounded intervals spectral methods converge at $O(e^{-cN})$. We therefore claim no demonstrated net-work advantage; the appeal of the construction is its explicit closed form and error structure.

5 Harmonic oscillator

For $y'' + \omega^2 y = 0$ with $y(0) = y_0$, $y'(0) = v_0$, define $C_n^{(s)}(x) := \Re E_n^{(s)}(ix)$ and $S_n^{(s)}(x) := \Im E_n^{(s)}(ix)$, and $y_n(t) = y_0 C_n^{(s)}(\omega t) + (v_0/\omega) S_n^{(s)}(\omega t)$.

Theorem 10 (Trigonometric properties). $C_n^{(s)} \rightarrow \cos$, $S_n^{(s)} \rightarrow \sin$ *uniformly on compacts*; $|C_{n+1}^{(s)} - C_n^{(s)}| = O(n^{-4})$ and likewise for S ; and the Pythagorean identity holds approximately: $(C_n^{(s)})^2 + (S_n^{(s)})^2 = 1 + O(n^{-3})$. *[num, classical]; base identity [Lean]*

Pythagorean identity. Since $(C_n^{(s)})^2 + (S_n^{(s)})^2 = |E_n^{(s)}(ix)|^2$ and $|E_n^{(s)}(ix)|^2 = \exp(2 \Re \log E_n^{(s)}(ix))$,

$$\begin{aligned} |E_n^{(s)}(ix)|^2 &= \exp(2 \Re(ix + g_3^{(s)}(ix)n^{-3} + O(n^{-4}))) \\ &= \exp(2 \Re g_3^{(s)}(ix) n^{-3} + O(n^{-4})) = 1 + 2 \Re g_3^{(s)}(ix) n^{-3} + O(n^{-4}), \end{aligned}$$

using Theorem 4(4). (This route avoids any claim that a complex power contributes a modulus-1 factor.) The polynomial base identity $|B_n^{(s)}(x)|^2 = 1 + x^2(1 - s\sqrt{3}/3)u^2 + (x^4/12)u^4$ is kernel-checked (Appendix 8, `pyth_expand`); it is consistent with the result but is not needed for it. $\square$

Lemma 11 (Derivative cancellation). $\frac{\partial}{\partial c} \log E_n^{(s)}(c) = 1 + O(n^{-3})$ *uniformly on compact c -sets*. Consequently $\frac{d}{dt} E_n^{(s)}(\omega t) = \omega e^{\omega t} + O(n^{-3})$. *[Lean] (the u^1 vanishing), [classical] (the u^2 vanishing, by differentiating the formalised A_2 identity), [num] (the $O(n^{-3})$)*

Proof. Differentiating $M(u) \log B(u)$ in c , the u^0 coefficient is 1 and both the u^1 and u^2 coefficients vanish: the u^1 coefficient is $2b_s + s\sqrt{3}/3 - 1$ with $b_s := 1/2 - s\sqrt{3}/6$, which is 0 for both branches (kernel-checked, `deriv_u1_plus/minus`), and the u^2 coefficient vanishes as well (it is the c -derivative of the A_2 polynomial identity, classical). Thus differentiation preserves the n^{-3} order (the error is analytic in c and u , so analytic differentiation on a compact set interior preserves the order). The derivative is therefore $1 + O(n^{-3})$, not merely $1 + O(n^{-2})$. $\square$

Proposition 12 (Energy conservation). With $E(t) = \frac{1}{2}(y')^2 + \frac{1}{2}\omega^2 y^2$ and $E_n(t)$ its telescoping analogue, $|E_n(t) - E(0)| = O(n^{-3})$. *[num]*

Proof. The derivative error is $O(n^{-3})$ (Lemma 11) and the Pythagorean error is $O(n^{-3})$ (Theorem 10); both contribute at that order. $\square$

6 Matrix differential equations

For $x' = Ax$, $x(0) = x_0$, with $A \in \mathbb{R}^{d \times d}$, put $X = At$ and define

$$M_n^{(s)}(t) = \left(I + \frac{X}{n} + s \frac{\sqrt{3}}{6} X^2 n^{-2}\right)^{nI + b_s X}, \quad b_s = \frac{1}{2} - s \frac{\sqrt{3}}{6},$$

the matrix power via $X^Y = \exp(Y \log X)$ (principal logarithm of the base).

Lemma 13 (Commutativity). $B_n^{(s)}(t) := I + \frac{X}{n} + s \frac{\sqrt{3}}{6} X^2 n^{-2}$ is a polynomial in X , hence commutes with X , with its (locally analytic) logarithm, and with the exponent $nI + b_s X$. [classical]

Theorem 14 (Matrix convergence). For any $A \in \mathbb{R}^{d \times d}$ and compact $I \subset \mathbb{R}$, for $n \geq n_0(A, I)$: $M_n^{(s)}(t) \rightarrow e^X$ in operator norm, $\|M_{n+1}^{(s)} - M_n^{(s)}\|_{\text{op}} = O(n^{-4})$, and $\|e^X - M_n^{(s)}\|_{\text{op}} = O(n^{-3})$, uniformly on I . [num]

Proof. This is the step the first draft asserted; here it is proved. Let $X = At$ and define the analytic logarithm

$$H_n = (nI + b_s X) \text{Log}\left(I + \frac{X}{n} + s \frac{\sqrt{3}}{6} X^2 n^{-2}\right),$$

so that $M_n^{(s)}(t) = \exp(H_n)$ by definition. By Lemma 13 every object in sight is a power series in the single matrix X , and these all commute. Hence H_n is the *scalar* function $M(u) \log B(u)$ of Section 3 evaluated at the commuting operator X (with c replaced by X): on a common eigenbasis, or block-by-block on a Jordan form, the u^1 and u^2 cancellations of Theorem 4 hold, giving

$$H_n = X + G_3(X) n^{-3} + O(n^{-4}), \quad G_3(X) = \frac{-3+2s\sqrt{3}}{72} X^4,$$

in operator norm, uniformly for $t \in I$. Because $[X, H_n - X] = 0$, $M_n = e^X \exp(H_n - X) = e^X + e^X G_3(X) n^{-3} + O(n^{-4})$, so $\|e^X - M_n^{(s)}\|_{\text{op}} = O(n^{-3})$; and $M_{n+1} - M_n = -3 e^X G_3(X) n^{-4} + O(n^{-5}) = O(n^{-4})$ by Lemma 5. The reduction is exact because the coefficient identities are polynomial identities in the commuting matrix X ; no commutator (Baker–Campbell–Hausdorff) terms arise. Note “ $\log M_n$ ” means H_n , the analytic logarithm, not necessarily the principal logarithm of the matrix $M_n = \exp H_n$ (these differ by $2\pi i$ multiples when eigenvalues leave the principal strip). $\square$

Remark 15. Verified numerically at 80-digit precision on a *non-normal* 3×3 matrix (driver, Part 3; $\|A^T A - A A^T\|_2 \approx 1.285$): $n^3 \|M_n - e^X\|_{\text{op}} \rightarrow 0.0011759 \dots$ (a stable constant, confirming $O(n^{-3})$) and $n^4 \|M_{n+1} - M_n\|_{\text{op}} \rightarrow 0.0035277 \dots$

Theorem 16 (Stability on compact time intervals). If all eigenvalues of A satisfy $\Re \lambda_i \leq -\gamma < 0$ and A is diagonalisable with $A = P D P^{-1}$, then for $n \geq n_0(A, T, \gamma)$, $\|M_n^{(s)}(t)\|_{\text{op}} \leq C(A) e^{-\gamma t}$ on $[0, T]$ with $C(A) = 2\|P\| \|P^{-1}\|$. [num]

Remark 17. We do *not* claim uniform-in-time A -stability; the bound is on a compact interval with n chosen for it. For non-diagonalisable A the result extends via Jordan form with polynomial factors absorbed into $C(A)$ (the same commuting-operator argument as Theorem 14).

7 General engine

Definition 18 (Exponential-algebraic class $\mathcal{G}$). $\mathcal{G}$ is the smallest class containing all $e^{\lambda t}$ ($\lambda \in \mathbb{C}$) and all polynomials t^k , closed under addition, multiplication, scalar multiplication, and differentiation. It is exactly the finite combinations $\sum c_{j,\lambda} t^j e^{\lambda t}$, i.e. the solution class of every constant-coefficient linear ODE [11].

Theorem 19 (ODE telescoping engine). *Let $L[y] = \sum_{i=0}^m a_i y^{(i)}$ have constant coefficients, and let y solve $L[y] = 0$ with prescribed initial data. If $y \in \mathcal{G}$ then there is a telescoping approximant y_n with $y_n \rightarrow y$ uniformly on compacts, $|y_{n+1} - y_n| = O(n^{-(k+1)})$, and $|y - y_n| = O(n^{-k})$, where k is the telescoping order of the underlying exponential approximants (here $k = 3$). **[num, classical]***

Proof. By the characteristic-polynomial structure, $y(t) = \sum_i \sum_{j < m_i} c_{i,j} t^j e^{\lambda_i t}$ [11]. Define $y_n(t) = \sum_{i,j} c_{i,j} t^j E_n^{(s)}(\lambda_i t)$, keeping the polynomial factor t^j *exact* and approximating only $e^{\lambda_i t}$. Each term converges with the required rate (Theorem 4); the finite sum preserves it by the triangle inequality. The telescoping difference bound follows the same way term-by-term. $\square$

Remark 20 (Differentiation and order). For this analytic family, differentiation *preserves* the telescoping order: Lemma 11 shows $\partial_c \log E_n^{(s)} = 1 + O(n^{-3})$ (both the u^1 and u^2 coefficients vanish), so the derivative is accurate to $O(n^{-3})$, the same order as the function. The engine’s conclusions are about the undifferentiated solution y_n ; the closure of $\mathcal{G}$ under differentiation holds at the same rate here, by analytic differentiation on the interior of a compact set.

8 Discussion

The framework’s appeal is analytical transparency: closed-form approximants with explicit error structure, at a polynomial rate $N = O(\varepsilon^{-1/3})$ that is slower than exponential-integrator or spectral methods [16, 19]. We make no net-cost claim. Natural extensions — higher-order kernels with $O(n^{-6})$ differences, hybrid schemes with modern exponential integrators [9, 18, 3, 2], and stiff neural-ODE contexts [13, 14] — are left open.

Declarations

Use of AI. The author acknowledges the use of a large-language-model assistant (Grok 4, xAI, December 2025) in developing an earlier draft: expanding and reorganising the author’s mathematical notes, drafting prose, and assisting literature search. *All mathematical content, proofs, and computations are the author’s.* The present revision (the corrected matrix proof, the differentiation/energy corrections, the reference corrections, and the Lean development) was also prepared with large-language-model assistance under the author’s direction. A known risk of AI-assisted reference finding is imperfect bibliographic metadata; all references were therefore checked against Crossref/arXiv/DataCite, and three author-name or institution errors in the earlier draft were corrected.

Evidence tiers. Claims are labelled **[Lean]** (kernel-checked in Appendix 8), **[num]** (confirmed by high-precision computation, not proved), and **[classical]** (standard facts). The asymptotic statements are *not* formalised.

Appendix A: Reproduction

The program `code/verify_ode.py` regenerates every numeral here (symbolic derivation of $g_1..g_5$; the derivative cancellation; the Pythagorean base; the leading difference constant by a Neville tableau; the high-precision non-normal matrix check; the derivative-order check). Appendix 8 ships the Lean development.

Appendix B: Lean development

The exact algebra is kernel-checked in Lean 4 (Mathlib), in twelve audited theorems. These include: the log-series coefficient identities `u1_coeff_zero` ($A_1 = 0$), `u2_coeff_zero` ($A_2 = 0$ given $(s\sqrt{3})^2 = 3$), and `u3_coeff_eq_g3` ($A_3 = g_3$); the finite-difference factor -3 (`diff_cube_factor`); the leading-constant identity $-3g_3^{(s)} = c^4(3 - 2s\sqrt{3})/24$ (`lead_eq`) and positivity of the $s = -1$ branch constant (`leadMinus_pos`); the derivative-cancellation vanishings (`deriv_u1_plus/minus`); and the Pythagorean base identity (`pyth_expand`). The gate `lean/gate.sh` requires a clean build, no `sorry/admit/axiom/native_decide`, only the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), and a minimum count of audited theorems; it prints `PASS (12 theorems, standard axioms only)`. The asymptotics ($O(n^{-3})$, $O(n^{-4})$, uniform convergence) are *not* formalised and are tiered [`num`].

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